

Project Title: Humpback Whale Identification.

Description and Goals:

The goal of this project is to identify a whale by its tail. The data can be found here: <https://www.kaggle.com/competitions/humpback-whale-identification/data>. The goal of this project is to build an algorithm to identify individual whales in over 25,000 images.

Responsibilities:

Team Member	Responsibilities
Kenia Sanchez-Macario	Model Training, Algorithm application
Jiarui (Lisa) Liu	Data Visualization; Correlation Analysis
Osvaldo Valdiviezo	Model Analysis; Preprocessing

Information about the data

Our datasets training data contains thousands of images of humpback whale flukes or tails. Individual whales have been identified by researchers and given an 'Id'. The challenge is to predict the whale 'Id' of images in the test set. The training set contains 9,850 images of humpback whales, with labels identifying the whale in each image. There are 5,004 unique classes, 25,361 training images and 7,960 test images which may increase as the dataset is increased over time. The dataset is purely image and identity label.

Regarding completeness, the dataset is partially clean but does have some disparities in the type of whale photographed. Much of the dataset is highly sparse for majority of whale identities. There are various duplicate images, class imbalance and inconsistencies across the dataset. This means we will need to preprocess the dataset before applying for training.

Feature Types:

- Image which is a nominal (string) containing the filename identifier with no ordinal or numeric meaning
- Id nominal (categorical) contains whale identity label of around 5,000+ discrete classes and 'new_whale'

Project status and progress so far:

The project has made steady progress. We started with exploration data analysis and understood that we must preprocess the datasets to gather a baseline of good dataset to train our model on. We are continuing now to use image classifiers to find the best accuracy on a best fit algorithm.